

Pratyay Dutta

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RESEARCH INTERESTS

Broad interest: Computer Vision, Deep Learning, Large Language Models (LLMs), Interpretability, Physics-based Modeling.

Specific interest: Physics-based modeling, Diffusion models, Person Re-Identification, Re-ranking, Occlusion detection, Uncertainty Quantification, Latent feature reconstruction, Heat diffusion transformer systems, Reinforcement learning, GNNs, Graph Attention, Data structures and algorithms, LLMs, VLMs, RLHF, DPO.

EDUCATION

University of California, Riverside

Doctor of Philosophy (Ph.D.) in Computer Science | GPA: 3.94/4.0

Aug 2023 – Present

California, USA

Jadavpur University

Bachelor of Engineering in Electronics and Telecommunication Engineering | GPA: 9.5/10

Jul 2019 – Aug 2023

Kolkata, India

Calcutta Boys' School

Primary, Middle and High School

Jul 2005 – Jul 2019

Kolkata, India

TECHNICAL SKILLS

Languages: Python, MATLAB, Java, C/C++, SQL

Developer Tools: Git, Docker, Google Cloud Platform, VS Code, PyCharm, Virtual environments, CUDA, Distributed training (DDP and FSDP)

Libraries: PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, Pandas, NumPy, Matplotlib

RESEARCH EXPERIENCE

Graduate Research Assistant

University of California, Riverside – Advisor: Dr. Bir Bhanu

Sep 2023 – Present

California, USA

- **Diffusion Priors for Efficient Occluded 3D Object Reconstruction:** Developed a robust diffusion prior method to generate multi-view images that minimizes the uncertainty of 3D Gaussian Splatting (3DGS) via Delta Denoising Score. Generated robust 3D meshes and objects from a single occluded image based on the diffusion prior.
- **Virtual Cloth Try-off using Stable Diffusion and Flux-Kontext:** Developed a virtual garment extraction system that maps images of clothed individuals to flat-lay representations of their clothing. Fine-tuned the denoising pipeline of **Stable Diffusion 1.5** on the VTON dataset and benchmarked it against two new vision-language models, **Flux-Kontext 1.0** (Black Forest Labs) and Google's **Nano-Banana**, demonstrating superior visual quality and semantic accuracy.
- **Image Inpainting with Diffusion Models:** Developed a robust pipeline to edit images using instance segmentation masks and diffusion models. Used bounding-box prompts with **SAM** and **DINO** for text-prompted segmentation of the editing region, and fine-tuned **Stable Diffusion 1.5** with ControlNet and **LoRA** to accurately inpaint the background after human body removal.
- **Rigid Body Collision Simulation:** Designed a **MuJoCo** simulator with custom physics-based modeling to prevent infinite bouncing, using a time integration scheme that handles collisions and contact separately. [\[Git\]](#)
- **Atari with Deep Recurrent Q-Networks:** Leveraged **baseline DQN with LSTM** to solve Atari games as POMDPs, outperforming standard DQN by leveraging temporal information and flow of information. [\[Git\]](#)
- **Efficient Multiple Sequence Alignment:** Implemented a Numba-accelerated **Hirschberg's algorithm** for 3-way MSA using space-saving dynamic programming, accelerating large NumPy matrix multiplications. [\[Git\]](#)

Data Science Summer Institute (DSSI) Graduate Student Intern

Lawrence Livermore National Laboratory

Jun 2026 – Aug 2026

California, USA

Advisors: Dr. Vivek Narayanaswamy, Dr. Kowshik Thopalli

- **COMPASS – Inference-Time Steering for LLM Reasoning:** Designed and implemented an activation-steering method that elicits chain-of-thought-quality reasoning from frozen LLMs using only the correctness of the model's own direct answers, by deriving a per-head correctness direction from attention-head activations, ranking heads with a logit-space attribution score, and steering the top-8 heads at inference time. Built the end-to-end **PyTorch/Hugging**

Face pipeline on multi-GPU SLURM clusters and evaluated **Llama-3.1-8B**, **Qwen3-4B**, and **Gemma-4-12B** on GSM8K, SVAMP, MATH-500, GSM-Plus, and HARP: improved GSM8K accuracy by 16 percentage points on average, outperformed the ITI and Fractional Reasoning baselines, and approached chain-of-thought accuracy with 20–70% fewer generated tokens, with interventions transferring to unseen benchmarks without re-fitting. Manuscript submitted to NeurIPS workshop : Interpretability as a Science.

- **DSSI Challenge Problem – Missing-Strut Detection in X-ray CT Scans of Additively Manufactured Lattices:** Built a registration-and-verification pipeline that detects missing struts in 1 GB X-ray computed tomography volumes of laser powder bed fusion Ti-5553 octet-truss lattices, registering the 10,206-junction nominal blueprint to each scan with a 7-DOF similarity fit (autocorrelation pitch estimation, junction-on-metal phase search, trimmed ICP) and verifying each of the 18,468 struts with a local matched-filter contrast test. Achieved 100% recall with zero false negatives across all 9 scans against STL design ground truth, with about 3 false positives per 13,000 struts on defect-free controls, and packaged the detector as a library, CLI, and Model Context Protocol (MCP) tool powering a FastAPI/React inspection dashboard with 3D defect visualization and LLM-assisted defect interpretation.

MITACS Globalink Research Intern

Jun 2022 – Aug 2022

Dalhousie University – Advisor: Dr. Yannick Marchand

Halifax, Canada

- Built a predictive modeling pipeline on **Lichess** chess data, identifying features critical to classifying problem difficulty for educational use, and built an RL agent using **PPO** that beats amateur players under a certain FIDE rating.

Research Intern

Jun 2021 – Nov 2021

Virginia Commonwealth University – Advisor: Dr. Preetam Ghosh

Virginia, USA

- Developed an optimized multi-vaccine distribution strategy based on demographic and heterogeneous factors in the Chicago district, using linear optimization (**PuLP**) and clustering algorithms. Published the resulting paper at IEEE BIBM 2021 and presented it virtually at the conference.
- Extended the scheme with hierarchical clustering and vaccine redistribution among zones for better allocation. Published the resulting journal paper in IEEE/ACM TCBB 2022.

PUBLICATIONS

- Pratyay Dutta, Ankith Jain Rakesh Kumar, Nghia Nguyen, Maziar Raissi, Bir Bhanu. “SAKE: Structure Aware Kernel Experts for Event-Based Object Detection,” *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*. (Major revision under way)
- Pratyay Dutta, Bir Bhanu. “Latent Diffusion-Guided Feature Inpainting for Occluded Person Re-Identification With Hybrid Re-Ranking,” *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, 2026. [\[doi\]](#)
- Pratyay Dutta, Padmaja Jonnalagedda, Bir Bhanu. “POANet: Parts-based Occlusion Aware Network for Person Re-ID,” *IEEE Transactions on Biometrics, Behavior, and Identity Science (TBIOM)*, 2026. [\[doi\]](#)
- Satyaki Roy, Pratyay Dutta, Preetam Ghosh. “Hierarchical Vaccine Allocation based on Epidemiological and Behavioral Considerations,” *IEEE/ACM TCBB 2022*. [\[doi\]](#)
- Satyaki Roy, Pratyay Dutta, Preetam Ghosh. “Generalizable multi-vaccine distribution strategy based on demographic and behavioral heterogeneity,” *IEEE BIBM 2021*. [\[doi\]](#)

HONORS AND AWARDS

- **Bishop Thoburn Award, 2019:** Student of the Year award from Calcutta Boys’ School (High School) for outstanding academic, sports and extra-curricular acumen (1 in 120 students).
- **MITACS Fellowship, 2020** for Summer Research Internship in Canada (Acceptance Rate: 18%).
- **Dean’s Fellowship, 2023** for outstanding research acumen (Acceptance Rate: 15%).

TEACHING AND PROFESSIONAL SERVICE

Teaching: CS170: Introduction to Artificial Intelligence, CS111: Discrete Structures, CS258: Introduction to Reinforcement Learning, EE152: Image Processing, EE114: Probability and Random Variables, CS105: Data Analysis Methods.

Reviewer: CVPR, WACV, ICCV, ECCV, NeurIPS, TBIOM, TCBB, TCSVT.